Contents

1 Mathematical Foundations of GAMLSS

This document provides detailed mathematical derivations for the GAMLSS implementation in `glissando`. It covers distribution-specific derivatives, special functions, numerical algorithms, and convergence theory.

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1.1 1. Distribution Derivatives

This section provides detailed derivations of score functions (first derivatives of log-likelihood) and Fisher information (expected second derivatives) for all implemented distributions. These quantities drive the penalized iteratively reweighted least squares (P-IRLS) algorithm.

Key quantities: - **Score** $u = \frac{\partial \ell}{\partial \eta}$: Direction of steepest ascent for the log-likelihood - **Fisher information** $w = I_\eta$: Expected curvature (used as weight in IRLS) - **Working response** $z = \eta + u/w$: Adjusted target for the weighted least squares problem

For each distribution, we derive these quantities accounting for the link function via the chain rule.

1.1.1 1.1 Gaussian Distribution

Parameterization: $Y \sim N(\mu, \sigma^2)$

Parameters: - μ (mean): identity link - σ (standard deviation): log link

Variance: $\text{Var}(Y) = \sigma^2$

1.1.1.1 Log-Likelihood

$$\ell(\mu, \sigma | y) = -\frac{1}{2}\log(2\pi) - \log(\sigma) - \frac{(y-\mu)^2}{2\sigma^2}$$

1.1.1.2 Derivatives for μ (identity link)

Score:

$$\frac{\partial \ell}{\partial \mu} = \frac{y-\mu}{\sigma^2}$$

Fisher information:

$$I_\mu = \mathbb{E}\left[-\frac{\partial^2 \ell}{\partial \mu^2}\right] = \frac{1}{\sigma^2}$$

Since we use identity link ($\eta = \mu$), no chain rule needed:

$$u_\mu = \frac{y-\mu}{\sigma^2}, \quad w_\mu = \frac{1}{\sigma^2}$$

1.1.1.3 Derivatives for σ (log link)

Let $\eta_\sigma = \log(\sigma)$, so $\sigma = e^{\eta_\sigma}$.

Score with respect to σ:

$$\frac{\partial \ell}{\partial \sigma} = -\frac{1}{\sigma} + \frac{(y-\mu)^2}{\sigma^3}$$

Chain rule for log link:

$$\frac{\partial \ell}{\partial \eta_\sigma} = \sigma \cdot \frac{\partial \ell}{\partial \sigma} = -1 + \frac{(y-\mu)^2}{\sigma^2} = \frac{(y-\mu)^2 - \sigma^2}{\sigma^2}$$

Fisher information on η_σ scale:

$$I_{\eta_\sigma} = \mathbb{E}\left[-\frac{\partial^2 \ell}{\partial \eta_\sigma^2}\right] = 2$$

This comes from $\mathbb{E}[(Y-\mu)^2/\sigma^2] = 1$ and $\mathrm{Var}[(Y-\mu)^2/\sigma^2] = 2$ for Gaussian.

Implementation:

$$u_\sigma = \frac{(y-\mu)^2 - \sigma^2}{\sigma^2}, \quad w_\sigma = 2$$

1.1.2 1.2 Student-t Distribution

1.1.3 Log-Likelihood

For observation y with parameters μ (location), σ (scale), and ν (degrees of freedom):

$$\ell(\mu, \sigma, \nu | y) = \log\Gamma\left(\frac{\nu+1}{2}\right) - \log\Gamma\left(\frac{\nu}{2}\right) - \frac{1}{2}\log(\nu\pi\sigma^2) - \frac{\nu+1}{2}\log\left(1 + \frac{z^2}{\nu}\right)$$

where $z = \frac{y-\mu}{\sigma}$.

1.1.4 First Derivatives (Score Functions)

Define the robustified weight:

$$w = \frac{\nu+1}{\nu+z^2}$$

1.1.4.1 Location parameter μ:

$$\frac{\partial \ell}{\partial \mu} = \frac{w \cdot z}{\sigma}$$

1.1.4.2 Scale parameter σ (log link):

$$\frac{\partial \ell}{\partial \log(\sigma)} = w \cdot z^2 - 1$$

1.1.4.3 Degrees of freedom ν (log link):

$$\frac{\partial \ell}{\partial \log(\nu)} = \nu \left[\frac{1}{2} \left(\psi\left(\frac{\nu+1}{2}\right) - \psi\left(\frac{\nu}{2}\right) - \log\left(1 + \frac{z^2}{\nu}\right) + \frac{w \cdot z^2 - 1}{\nu} \right) \right]$$

where $\psi(x) = \frac{d}{dx} \log \Gamma(x)$ is the digamma function.

1.1.5 Expected Fisher Information

1.1.5.1 For μ:

$$I_\mu = \mathbb{E}\left[\frac{w}{\sigma^2}\right] = \frac{\nu+1}{\sigma^2(\nu+3)}$$

In practice, we use the observed information (plug in w directly).

1.1.5.2 For σ (log link):

$$I_{\log(\sigma)} = \frac{2\nu}{\nu+3}$$

1.1.5.3 For ν (log link):

$$I_{\log(\nu)} = \frac{\nu^2}{4} \left[\psi'\left(\frac{\nu}{2}\right) - \psi'\left(\frac{\nu+1}{2}\right) + \frac{2(\nu+3)}{\nu(\nu+1)} \right]$$

where $\psi'(x)$ is the trigamma function.

1.1.6 Numerical Considerations

1. **Minimum ν**: Require $\nu > 2$ to ensure finite variance
2. **Minimum σ**: Enforce $\sigma \geq 10^{-6}$ to prevent division by zero
3. **Weight clamping**: The denominator $\nu + z^2$ should be $\geq 10^{-10}$
4. **Information positivity**: Ensure $I_{\log(\nu)} \geq 10^{-6}$

1.1.7 1.3 Poisson Distribution

Parameterization: $Y \sim \text{Poisson}(\mu)$

Parameters: - μ (rate/mean): log link

Variance: $\text{Var}(Y) = \mu$ (equidispersion)

1.1.7.1 Log-Likelihood

$$\ell(\mu|y) = y \log(\mu) - \mu - \log(y!)$$

1.1.7.2 Derivatives for μ (log link)

Let $\eta = \log(\mu)$, so $\mu = e^\eta$.

Score with respect to μ:

$$\frac{\partial \ell}{\partial \mu} = \frac{y}{\mu} - 1 = \frac{y - \mu}{\mu}$$

Chain rule for log link:

$$\frac{\partial \ell}{\partial \eta} = \mu \cdot \frac{\partial \ell}{\partial \mu} = y - \mu$$

Fisher information on η scale:

$$I_\eta = \mathbb{E}\left[-\frac{\partial^2 \ell}{\partial \eta^2}\right] = \mu$$

Implementation:

$$u_\mu = y - \mu, \quad w_\mu = \mu$$

1.1.8 1.4 Gamma Distribution

Parameterization: $Y \sim \text{Gamma}(\alpha, \theta)$ where $\alpha = 1/\sigma^2$ (shape), $\theta = \mu\sigma^2$ (scale)

Parameters: - μ (mean): log link - σ (coefficient of variation): log link

Variance: $\text{Var}(Y) = \mu^2\sigma^2$

Mean-CV relationship: $\text{CV}(Y) = \sigma$

1.1.8.1 Log-Likelihood

$$\ell(\mu, \sigma | y) = -\alpha \log(\theta) - \log\Gamma(\alpha) + (\alpha - 1)\log(y) - \frac{y}{\theta}$$

where $\alpha = 1/\sigma^2$ and $\theta = \mu\sigma^2$.

Simplifying:

$$\ell = -\frac{1}{\sigma^2}\log(\mu\sigma^2) - \log\Gamma\left(\frac{1}{\sigma^2}\right) + \left(\frac{1}{\sigma^2} - 1\right)\log(y) - \frac{y}{\mu\sigma^2}$$

1.1.8.2 Derivatives for μ (log link)

Let $\eta_\mu = \log(\mu)$.

Score with respect to μ:

$$\frac{\partial \ell}{\partial \mu} = -\frac{\alpha}{\mu} + \frac{y}{\mu^2\theta} = -\frac{1}{\mu\sigma^2} + \frac{y}{\mu^2\sigma^2} = \frac{y - \mu}{\mu^2\sigma^2}$$

Chain rule for log link:

$$\frac{\partial \ell}{\partial \eta_\mu} = \mu \cdot \frac{\partial \ell}{\partial \mu} = \frac{y - \mu}{\mu\sigma^2}$$

Fisher information on η_μ scale:

$$I_{\eta_\mu} = \frac{1}{\sigma^2}$$

Implementation:

$$u_\mu = \frac{y - \mu}{\mu\sigma^2}, \quad w_\mu = \frac{1}{\sigma^2}$$

1.1.8.3 Derivatives for σ (log link)

Let $\eta_\sigma = \log(\sigma)$.

Score with respect to σ:

$$\frac{\partial \ell}{\partial \sigma} = -\frac{2\alpha}{\sigma} + \frac{2\alpha}{\sigma}\psi(\alpha) + \frac{2\alpha}{\sigma}\log(y) - \frac{2\alpha}{\sigma}\log(\theta) + \frac{2y}{\mu\sigma^3}$$

where $\psi(x) = \frac{d}{dx}\log\Gamma(x)$ is the digamma function.

Chain rule for log link ($\frac{d\sigma}{d\eta_\sigma} = \sigma$):

$$\frac{\partial \ell}{\partial \eta_\sigma} = \frac{2}{\sigma^2}\left[\psi\left(\frac{1}{\sigma^2}\right) + 2\log(\sigma) - \log\left(\frac{y}{\mu}\right) + \frac{y}{\mu} - 1\right]$$

Fisher information involves trigamma $\psi'(x) = \frac{d^2}{dx^2}\log\Gamma(x)$:

$$I_{\eta_\sigma} = \frac{4}{\sigma^4}\psi'\left(\frac{1}{\sigma^2}\right) - \frac{2}{\sigma^2}$$

Implementation:

$$u_\sigma = \frac{2}{\sigma^2}\left[\psi(\alpha) + 2\log(\sigma) - \log(y/\mu) + y/\mu - 1\right], \quad w_\sigma = \frac{4}{\sigma^4}\psi'(\alpha) - \frac{2}{\sigma^2}$$

1.1.9 1.5 Negative Binomial Distribution

Parameterization: $Y \sim \text{NB}(r, p)$ with mean μ and dispersion σ (NB2 parameterization)

Parameters: - μ (mean): log link - σ (overdispersion): log link

Variance: $\text{Var}(Y) = \mu + \sigma\mu^2$ (overdispersion relative to Poisson)

Relationship to standard NB: $r = 1/\sigma$, $p = 1/(1+\sigma\mu)$

1.1.9.1 Log-Likelihood

$$\ell(\mu, \sigma | y) = \log\Gamma(y + r) - \log\Gamma(r) - \log(y!) + r\log(p) + y\log(1-p)$$

where $r = 1/\sigma$ and $p = 1/(1+\sigma\mu)$.

Simplifying in terms of (μ, σ):

$$\ell = \log\Gamma(y + 1/\sigma) - \log\Gamma(1/\sigma) - \log(y!) + \frac{1}{\sigma}\log\left(\frac{1}{1+\sigma\mu}\right) + y\log\left(\frac{\sigma\mu}{1+\sigma\mu}\right)$$

1.1.9.2 Derivatives for μ (log link)

Let $\eta_\mu = \log(\mu)$.

Score with respect to μ:

$$\frac{\partial \ell}{\partial \mu} = \frac{y - \mu}{1 + \sigma\mu}$$

Chain rule for log link ($\frac{d\mu}{d\eta_\mu} = \mu$):

$$\frac{\partial \ell}{\partial \eta_\mu} = \mu \cdot \frac{\partial \ell}{\partial \mu} = \frac{\mu(y-\mu)}{1+\sigma\mu}$$

However, in IRLS we use the observed Fisher information $w_\mu = \frac{\mu}{1+\sigma\mu}$, giving working response:

$$u_\mu = \frac{y-\mu}{1+\sigma\mu}, \quad w_\mu = \frac{\mu}{1+\sigma\mu}$$

This follows from the score being proportional to observed weight, as noted in the full derivation in Rigby & Stasinopoulos (2005).

Why observed information here: for μ in NB the expected information $I_{\eta_\mu} = \mathbb{E}[\mu(Y-\mu)/(1+\sigma\mu)^2]$ collapses to $\mu/(1+\sigma\mu)$ after expectation, which coincides numerically with the observed-information form because of the variance identity $\mathrm{Var}(Y) = \mu(1 + \sigma\mu)$. Rigby & Stasinopoulos (2005, eq. (15)) use this simplification; `src/distributions/negative_binomial.rs` follows it. As $\sigma \to 0$ the weight reduces to μ — exactly the Poisson Fisher weight — confirming the limiting Poisson behavior.

Implementation (for IRLS):

$$u_\mu = \frac{y - \mu}{1 + \sigma\mu}, \quad w_\mu = \frac{\mu}{1 + \sigma\mu}$$

1.1.9.3 Derivatives for σ (log link)

Let $\eta_\sigma = \log(\sigma)$ and $r = 1/\sigma$.

Score with respect to σ:

$$\frac{\partial \ell}{\partial \sigma} = -\frac{1}{\sigma^2}\left[\psi(y + r) - \psi(r) - \log(1 + \sigma\mu) + \frac{y - \mu}{1 + \sigma\mu}\right]$$

Chain rule for log link:

$$\frac{\partial \ell}{\partial \eta_\sigma} = \sigma \cdot \frac{\partial \ell}{\partial \sigma} = -\frac{1}{\sigma}\left[\psi(y + r) - \psi(r) - \log(1 + \sigma\mu) + \frac{y - \mu}{1 + \sigma\mu}\right]$$

Fisher information (using approximation):

$$I_{\eta_\sigma} = \frac{1}{\sigma^2}\psi'(r)$$

Implementation:

$$u_\sigma = -\frac{1}{\sigma}\left[\psi(y + 1/\sigma) - \psi(1/\sigma) - \log(1 + \sigma\mu) + \frac{y - \mu}{1 + \sigma\mu}\right], \quad w_\sigma = \frac{\psi'(1/\sigma)}{\sigma^2}$$

1.1.10 1.6 Beta Distribution

Parameterization: $Y \sim \mathrm{Beta}(\alpha, \beta)$ where $\alpha = \mu\phi$ and $\beta = (1 - \mu)\phi$

Parameters: - μ (mean): logit link - ϕ (precision): log link

Variance: $\mathrm{Var}(Y) = \frac{\mu(1-\mu)}{1+\phi}$

1.1.10.1 Log-Likelihood

$$\ell(\mu, \phi | y) = \log\Gamma(\phi) - \log\Gamma(\alpha) - \log\Gamma(\beta) + (\alpha - 1)\log(y) + (\beta - 1)\log(1 - y)$$

where $\alpha = \mu\phi$ and $\beta = (1 - \mu)\phi$.

1.1.10.2 Derivatives for μ (logit link)

Let $\eta_\mu = \mathrm{logit}(\mu) = \log(\mu/(1 - \mu))$.

Score with respect to μ:

$$\frac{\partial \ell}{\partial \mu} = \phi\left[\log(y) - \log(1 - y) - \psi(\alpha) + \psi(\beta)\right]$$

Chain rule for logit link ($\frac{d\mu}{d\eta_\mu} = \mu(1 - \mu)$):

$$\frac{\partial \ell}{\partial \eta_\mu} = \mu(1 - \mu) \cdot \frac{\partial \ell}{\partial \mu} = \mu(1 - \mu)\phi\left[\log(y) - \log(1 - y) - \psi(\alpha) + \psi(\beta)\right]$$

Fisher information:

$$I_{\eta_\mu} = [\mu(1-\mu)]^2\phi^2[\psi'(\alpha) + \psi'(\beta)]$$

Implementation:

$$u_\mu = \mu(1-\mu)\phi\left[\log(y) - \log(1-y) - \psi(\mu\phi) + \psi((1-\mu)\phi)\right]$$
$$w_\mu = [\mu(1-\mu)]^2\phi^2[\psi'(\mu\phi) + \psi'((1-\mu)\phi)]$$

1.1.10.3 Derivatives for ϕ (log link) Let $\eta_\phi = \log(\phi)$.

Score with respect to ϕ:

$$\frac{\partial \ell}{\partial \phi} = \psi(\phi) - \mu\psi(\alpha) - (1-\mu)\psi(\beta) + \mu\log(y) + (1-\mu)\log(1-y)$$

Chain rule for log link:

$$\frac{\partial \ell}{\partial \eta_\phi} = \phi \cdot \frac{\partial \ell}{\partial \phi}$$

Fisher information:

$$I_{\eta_\phi} = \phi^2\left[\psi'(\phi) - \mu^2\psi'(\alpha) - (1-\mu)^2\psi'(\beta)\right]$$

Implementation:

$$u_\phi = \phi\left[\psi(\phi) - \mu\psi(\mu\phi) - (1-\mu)\psi((1-\mu)\phi) + \mu\log(y) + (1-\mu)\log(1-y)\right]$$
$$w_\phi = \phi^2\left[\psi'(\phi) - \mu^2\psi'(\mu\phi) - (1-\mu)^2\psi'((1-\mu)\phi)\right]$$

1.1.11 1.7 Binomial Distribution

Parameterization: $Y \sim \text{Binomial}(n, \mu)$ where Y is the number of successes in n trials

Parameters: - μ (probability): logit link

Variance: $\text{Var}(Y) = n\mu(1-\mu)$

1.1.11.1 Log-Likelihood

$$\ell(\mu|y) = y\log(\mu) + (n-y)\log(1-\mu) + \log\binom{n}{y}$$

1.1.11.2 Derivatives for μ (logit link) Let $\eta = \text{logit}(\mu)$.

Score with respect to μ:

$$\frac{\partial \ell}{\partial \mu} = \frac{y}{\mu} - \frac{n-y}{1-\mu} = \frac{y-n\mu}{\mu(1-\mu)}$$

Chain rule for logit link ($\frac{d\mu}{d\eta} = \mu(1-\mu)$):

$$\frac{\partial \ell}{\partial \eta} = \mu(1-\mu) \cdot \frac{\partial \ell}{\partial \mu} = y - n\mu$$

Fisher information on η scale:

$$I_\eta = n\mu(1-\mu)$$

Implementation:

$$u_\mu = y - n\mu, \quad w_\mu = n\mu(1-\mu)$$

1.1.12 1.8 Example: Computing Derivatives for Gaussian Data

Suppose we have data $y = [2.1, 1.9, 3.2]$ and current parameter estimates $\mu = [2.0, 2.0, 3.0]$, $\sigma = [0.5, 0.5, 0.5]$.

For μ (identity link):

Residuals: $r = y - \mu = [0.1, -0.1, 0.2]$

Weights: $w_\mu = 1/\sigma^2 = [4.0, 4.0, 4.0]$

Score: $u_\mu = r \cdot w_\mu = [0.4, -0.4, 0.8]$

Working response: $z_\mu = \mu + u_\mu / w_\mu = \mu + r = y$

(For identity link with constant weights, the working response is just y)

For σ (log link):

Squared residuals: $r^2 = [0.01, 0.01, 0.04]$

Score on $\eta_\sigma = \log(\sigma)$ scale:

$$u_\sigma = \frac{r^2 - \sigma^2}{\sigma^2} = \frac{[0.01, 0.01, 0.04] - 0.25}{0.25} = [-0.96, -0.96, -0.84]$$

Weights: $w_\sigma = [2.0, 2.0, 2.0]$

Working response on log scale:

$$z_\sigma = \log(\sigma) + u_\sigma / w_\sigma = [-0.693, -0.693, -0.693] + [-0.48, -0.48, -0.42] = [-1.17, -1.17, -1.11]$$

The PWLS solver then regresses z_σ on the design matrix for σ with weights w_σ to update σ.

1.2 2. Special Functions

1.2.1 2.1 Digamma Function

The digamma function $\psi(x) = \frac{d}{dx} \log \Gamma(x) = \frac{\Gamma'(x)}{\Gamma(x)}$ appears in derivatives for Gamma, Negative Binomial, and Beta distributions.

1.2.1.1 Recurrence Relation

For $x < 5$, use:

$$\psi(x) = \psi(x+1) - \frac{1}{x}$$

Repeatedly apply until $x \geq 5$.

1.2.1.2 Asymptotic Expansion

For $x \geq 5$, use the series (Abramowitz & Stegun 6.3.18):

$$\psi(x) \sim \log(x) - \frac{1}{2x} - \frac{1}{12x^2} + \frac{1}{120x^4} - \frac{1}{252x^6} + O(x^{-8})$$

1.2.1.3 Known Values (for testing)

- $\psi(1) = -\gamma \approx -0.5772156649$ (Euler-Mascheroni constant)
- $\psi(2) = 1 - \gamma \approx 0.4227843351$
- $\psi(1/2) = -\gamma - 2\log(2) \approx -1.9635100260$

1.2.2 2.2 Trigamma Function

The trigamma function $\psi'(x) = \frac{d^2}{dx^2} \log \Gamma(x)$ is needed for Student-t Fisher information.

1.2.3 Recurrence Relation

For $x < 5$, use:

$$\psi'(x) = \psi'(x+1) + \frac{1}{x^2}$$

1.2.4 Asymptotic Expansion

For $x \geq 5$, use the series (Abramowitz & Stegun 6.4.11):

$$\psi'(x) \sim \frac{1}{x} + \frac{1}{2x^2} + \frac{1}{6x^3} - \frac{1}{30x^5} + \frac{1}{42x^7} - \frac{1}{30x^9} + O(x^{-11})$$

1.2.5 Known Values (for testing)

- $\psi'(1) = \frac{\pi^2}{6} \approx 1.644934067$
- $\psi'(2) = \frac{\pi^2}{6} - 1 \approx 0.644934067$
- $\psi'(1/2) = \frac{\pi^2}{2} \approx 4.934802201$

1.3 3. Numerical Stability and Implementation

1.3.1 3.1 Parameter Bounds

To prevent numerical issues, parameters are clamped to safe ranges:

Minimum positive value: 10^{-10} - Used for: μ (when must be positive), σ, ϕ, probabilities

Minimum weight: 10^{-6} - Fisher information weights are floored at this value to ensure positive definiteness of the weight matrix

Link function bounds: - Log/logit links: $\eta \in [-30, 30]$ - Prevents overflow in $\exp(\eta)$ (since $e^{30} \approx 10^{13}$ and $e^{-30} \approx 10^{-14}$)

1.3.2 3.2 Distribution-Specific Safeguards

Student-t: - Require $\nu > 2$ to ensure finite variance - Ensure $\nu + z^2 \geq 10^{-10}$ to prevent division issues in robustifying weight

Gamma: - Clamp σ to reasonable range to prevent extreme shape parameters

Beta: - Clamp y to $(10^{-10}, 1 - 10^{-10})$ before computing $\log(y)$ and $\log(1-y)$ - Clamp μ similarly before computing logit

Negative Binomial: - Ensure $1 + \sigma\mu$ stays positive and bounded away from zero

1.3.3 3.3 Batched Special Function Computation

Digamma and trigamma functions are computed in batched vectorized form for performance:

- For $n <$ 10,000 observations: sequential computation
- For $n \geq$ 10,000: parallel computation via Rayon (when `parallel` feature enabled)

The switchpoint is empirically tuned to balance overhead vs. speedup.

1.4 4. GAMLSS Iteration (P-IRLS)

1.4.1 4.1 Backfitting Outer Loop

GAMLSS fits the joint model by cycling through distribution parameters one at a time, holding the others fixed (Rigby & Stasinopoulos 2005, §3). For a P-parameter family (e.g. Gaussian: $P = 2$; Student-t: $P = 3$):

```
for cycle in 0..max_iterations:
    max_diff = 0
    for each parameter theta_k in family.parameters():
        snapshot beta_k_old
        beta_k = p_irls_step(theta_k | other params held fixed)
        max_diff = max(max_diff, max_abs(beta_k - beta_k_old))
    if max_diff < tolerance:
        break
```

Source: `src/fitting/mod.rs:223-263`. The convergence criterion is the **infinity norm** of coefficient changes (max absolute element), pooled across all parameters in a cycle — not the L^1 norm of each parameter separately. The active default is $\epsilon = 10^{-3}$, max iterations = 200 (`DEFAULT_TOLERANCE`, `DEFAULT_MAX_ITER` at `src/fitting/mod.rs:31-32`).

1.4.2 4.2 Inner P-IRLS Step (Fisher Scoring)

For each parameter θ_k (e.g., μ, σ, ν):

1. **Working response**:
$$z_k = \eta_k + \frac{u_k}{w_k}$$
   where $u_k = \frac{\partial \ell}{\partial \eta_k}$ and w_k is the (expected, or in some cases observed) Fisher information.
2. **Penalized weighted least squares** (see §8.1 for the Cholesky solve):
$$\hat{\beta}_k = \arg\min_{\beta} \|W_k^{1/2}(z_k - X_k\beta)\|^2 + \sum_j \lambda_j \beta^T S_j \beta$$
3. **Update linear predictor**:
$$\eta_k^{(\text{new})} = X_k \hat{\beta}_k$$
4. **Smoothing parameter selection** — GCV, REML, or Fellner–Schall (§§ 8, 8.2).

Source: `src/fitting/scoring.rs:40-180`.

1.4.3 4.3 Working-Response Derivation

Why is $z = \eta + u/w$ the right pseudo-response? Expand the log-likelihood for one observation in η around the current iterate η_0:

$$\ell(\eta) \approx \ell(\eta_0) + u(\eta_0)\,(\eta - \eta_0) - \tfrac{1}{2}\,w(\eta_0)\,(\eta - \eta_0)^2$$

where $u = \partial\ell/\partial\eta$ and $w = -\mathbb{E}[\partial^2\ell/\partial\eta^2]$ (Fisher information, positive). Maximizing the quadratic Taylor surrogate over η gives the one-Newton-step update $\eta - \eta_0 = u/w$. Setting $\eta = X\beta$ and stacking observations yields the Gauss–Newton normal equations:

$$X^T W X\,\Delta\beta = X^T W\,(u/w)$$

with $\Delta\beta = \beta - \beta_0$ and the diagonal weight $W = \mathrm{diag}(w_i)$. Equivalently, regress the **working response** $z = \eta_0 + u/w$ on X with weights W:

$$X^T W X \beta = X^T W z.$$

Adding the penalty $\sum_j \lambda_j \beta^T S_j \beta$ to the surrogate produces the PWLS system of §8.1. This is exactly the IRLS construction of McCullagh & Nelder (1989) generalized to one distributional parameter at a time.

1.4.4 4.4 Numerical Safeguards in the Inner Loop

Cataloged here so the magic constants are auditable. All apply per observation, per inner iteration. Source: `src/fitting/scoring.rs`, `src/distributions/links.rs`.

Symbol	Value	Where	Purpose
`MIN_WEIGHT`	10^{-6}	`scoring.rs:28`	Floor on w_i so the weight matrix stays positive definite; near-zero Fisher info would blow up u/w. Hits counted in `weight_floor_hits`.
`MAX_STEP`	20.0	`scoring.rs:32`	Clamps u_i/w_i to ± 20 in η-units. Prevents wild Newton steps from outliers. Hits counted in `step_cap_hits`.
`MAX_ETA`, `MIN_ETA`	± 30	`links.rs:10-12`	Clamps η before applying the inverse log/logit link so $\exp(\eta)$ stays finite ($e^{30} \approx 10^{13}$).
`MIN_POSITIVE`	10^{-10}	`links.rs:8`, `diagnostics.rs:15`	Floor on parameters that must be strictly positive (σ, ϕ, probabilities), and on variances before the square-root in residual computation.

Persistent non-zero `weight_floor_hits` or `step_cap_hits` at convergence signals model misspecification or ill-conditioned data; transient hits in early iterations are normal.

1.5 5. Smoothing and Penalty Matrices

1.5.1 5.0 B-Spline Basis (Cox–de Boor)

A degree-p B-spline basis with k functions is defined on a clamped knot vector

$$\tau_0 = \tau_1 = \cdots = \tau_p < \tau_{p+1} < \cdots < \tau_{k-1} < \tau_k = \cdots = \tau_{k+p}.$$

The implementation places the $k-p-1$ interior knots at empirical **quantiles** of the predictor:

$$\tau_{p+1+i} = x_{(\,\mathrm{round}(\,q_i\,(n-1)\,))}, \qquad q_i = \frac{i}{k-p}, \quad i = 1, \dots, k-p-1$$

where $x_{(j)}$ is the j-th order statistic. Boundary knots are replicated $p+1$ times at $\min x$ and $\max x$ (`src/splines.rs:99-128`).

The **Cox–de Boor recursion** evaluates the basis stably:

$$B_{i,0}(x) = \begin{cases} 1 & \tau_i \le x < \tau_{i+1} \\ 0 & \text{otherwise} \end{cases}$$

$$B_{i,p}(x) = \frac{x - \tau_i}{\tau_{i+p} - \tau_i} B_{i,p-1}(x) + \frac{\tau_{i+p+1} - x}{\tau_{i+p+1} - \tau_{i+1}} B_{i+1,p-1}(x).$$

Properties:

- **Partition of unity**: $\sum_{j=1}^{k} B_j(x_i) = 1$ for every x_i in the interior of the knot range.
- **Non-negativity**: $B_j(x) \ge 0$.
- **Local support**: $B_{i,p}$ is non-zero only on $[\tau_i, \tau_{i+p+1}]$, so each row of the design matrix has at most $p+1$ non-zero entries.

Source: `src/splines.rs:42-160` (`create_basis_matrix`, `evaluate_basis_functions_into`, `find_knot_span` using `partition_point` for binary search).

1.5.2 5.1 Sum-to-Zero Reparameterization (Identifiability)

Partition-of-unity creates a rank deficiency when the model has both an intercept and a smooth term: the column vector $\mathbf{1}_n$ lies in the column space of B, so the design matrix $[\mathbf{1}_n \mid B]$ is column-rank-deficient. To restore identifiability, the smooth is reparameterized through an orthonormal basis of the subspace orthogonal to $\mathbf{1}_k$.

The implementation uses a **Householder reflector** (`src/splines.rs:1-39`). Choose

$$v = \mathbf{1}_k + \sqrt{k}\, e_1$$

which reflects $\mathbf{1}_k$ onto $-\sqrt{k}\, e_1$. Form

$$H = I_k - \frac{2}{\|v\|^2} vv^T, \qquad Z = H_{:,2:k}$$

i.e. drop the first column of H. Then $Z \in \mathbb{R}^{k \times (k-1)}$ satisfies

$$Z^T Z = I_{k-1}, \qquad Z^T \mathbf{1}_k = 0.$$

Reparameterize $\beta = Z\gamma$ to get the constrained basis and penalty:

$$B_{\text{new}} = BZ, \qquad S_{\text{new}} = Z^T S\, Z.$$

The new design matrix $[\mathbf{1}_n \mid BZ]$ has full column rank, and the constraint $\mathbf{1}_k^T \beta = 0$ — i.e. the smooth has mean zero across knots — is enforced automatically.

1.5.3 P-Spline Penalty (Eilers & Marx 1996)

For 2nd-order differences:

$$S = D^T D$$

where D is the $(m-2) \times m$ difference matrix:

$$D = \begin{bmatrix} 1 & -2 & 1 & 0 & \cdots & 0 \\ 0 & 1 & -2 & 1 & \cdots & 0 \\ \vdots & & \ddots & & & \vdots \\ 0 & \cdots & 0 & 1 & -2 & 1 \end{bmatrix}$$

This penalizes curvature: $\lambda \beta^T S \beta \approx \lambda \int (\beta''(x))^2 dx$

1.5.4 Tensor Product Smooth and Anisotropic Penalty

For a bivariate smooth $f(x_1, x_2)$ built from marginal bases $B_1 \in \mathbb{R}^{n\times k_1}$ and $B_2 \in \mathbb{R}^{n\times k_2}$, the joint basis is the **row-wise Kronecker product** of the marginals:

$$B[i,:] = B_1[i,:] \otimes B_2[i,:] \in \mathbb{R}^{k_1 k_2}, \qquad B \in \mathbb{R}^{n\times(k_1 k_2)}.$$

The coefficient vector $\beta \in \mathbb{R}^{k_1 k_2}$ is indexed lexicographically so $\beta_{(i-1)k_2+j}$ multiplies $B_1[\cdot, i] \cdot B_2[\cdot, j]$.

Anisotropic penalties are obtained by Kronecker'ing each marginal difference penalty with an identity matrix:

$$S^{(1)} = S_1 \otimes I_{k_2}, \qquad S^{(2)} = I_{k_1} \otimes S_2$$

with two independent smoothing parameters:

$$\lambda_1 \beta^T S^{(1)} \beta + \lambda_2 \beta^T S^{(2)} \beta.$$

$S^{(1)}$ penalizes roughness in the x_1 direction at every x_2 slice; $S^{(2)}$ vice versa. Choosing two λ's lets the GCV/REML routine pick different smoothness for each margin (Wood 2017, §5.6).

The sum-to-zero constraint of §5.1 is applied **per margin** before the Kronecker product when an intercept is present:

$$B_{1,\text{new}} = B_1 Z_1, \;\; B_{2,\text{new}} = B_2 Z_2, \;\; S^{(1)} = (Z_1^T S_1 Z_1) \otimes I_{k_2-1}, \;\; S^{(2)} = I_{k_1-1} \otimes (Z_2^T S_2 Z_2).$$

Source: `src/fitting/assembler.rs:44-102`, `src/splines.rs:8-41`.

1.5.5 Random Effect Penalty

For an observation-to-group map $g : \{1, \dots, n\} \to \{1, \dots, G\}$, the design matrix is the binary indicator

$$Z \in \{0,1\}^{n\times G}, \qquad Z[i,g] = \mathbf{1}\{g(i) = g\}.$$

Coupled with the **identity penalty** $S = I_G$, the model

$$\eta = \mathbf{1}\beta_0 + Z\alpha, \qquad \lambda \alpha^T \alpha = \lambda \sum_{g=1}^{G} \alpha_g^2$$

is equivalent to the Bayesian random-intercept prior $\alpha_g \sim N(0, 1/\lambda)$ — the ridge-penalized empirical Bayes interpretation. Since each row of Z sums to 1, the sum-to-zero reparameterization of §5.1 is again applied when an intercept is present:

$$Z_{\text{new}} = Z\, Z_{\text{c}} \in \mathbb{R}^{n\times(G-1)}, \qquad Z_{\text{c}}^T I_G Z_{\text{c}} = I_{G-1}.$$

Source: `src/fitting/assembler.rs:103-126`.

1.5.6 5.1 Block-Sparse Penalty Matrices

In practice, penalty matrices S_j are often **block-sparse**: they only penalize a subset of the coefficient vector. For example, if the model has an intercept, linear terms, and a smooth term:

$$\beta = [\beta_{\text{intercept}}, \beta_{\text{linear}}, \beta_{\text{smooth}}]^T$$

The penalty for the smooth term is:

$$S_{\text{smooth}} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & S_{\text{block}} \end{bmatrix}$$

where S_{block} is the $(n_{\text{splines}} \times n_{\text{splines}})$ penalty for the spline coefficients.

Efficient Storage: Instead of storing the full $p \times p$ matrix (mostly zeros), we store: - `block`: The non-zero sub-matrix S_{block} - `offset`: Starting index in the full coefficient vector - `full_dim`: Total dimension p

Operations:

Scaled addition: $A \leftarrow A + \lambda S$

$$A[i:i+k, j:j+k] \leftarrow A[i:i+k, j:j+k] + \lambda \cdot S_{\text{block}}$$

where $i = \text{offset}$ and $k = \dim(S_{\text{block}})$.

Matrix-vector product: $v = S\beta$

$$v[i:i+k] = S_{\text{block}} \cdot \beta[i:i+k], \quad v[\text{elsewhere}] = 0$$

This exploits sparsity for computational efficiency, especially important when many terms have independent penalties.

1.6 6. Effective Degrees of Freedom

The "wiggliness" of the fit is measured by:

$$\text{EDF} = \text{tr}(H)$$

where $H = X(X^T W X + \lambda S)^{-1} X^T W X$ is the hat matrix.

Interpretation: - EDF ≈ 2 for a linear fit (intercept + slope) - EDF $\approx$ number of basis functions for an unpenalized spline - EDF between these extremes reflects smoothing

---

1.7 7. Link Functions

1.7.1 Identity Link

- **Function**: $g(\mu) = \mu$
- **Inverse**: $g^{-1}(\eta) = \eta$
- **Domain**: $\mu \in \mathbb{R}$
- **Use**: Gaussian mean, Student-t location
- **Derivative**: $\frac{d\mu}{d\eta} = 1$

1.7.2 Log Link

- **Function**: $g(\mu) = \log(\mu)$
- **Inverse**: $g^{-1}(\eta) = \exp(\eta)$
- **Domain**: $\mu > 0$
- **Use**: Poisson rate, Gamma mean, scale parameters, degrees of freedom
- **Derivative**: $\frac{d\mu}{d\eta} = \mu$
- **Numerical safeguard**: Clamp $\eta \in [-30, 30]$ to prevent overflow

1.7.3 Logit Link

- **Function**: $g(\mu) = \log\left(\frac{\mu}{1-\mu}\right)$
- **Inverse**: $g^{-1}(\eta) = \frac{1}{1+e^{-\eta}} = \frac{e^\eta}{1+e^\eta}$
- **Domain**: $\mu \in (0, 1)$

- **Use**: Binomial probability, Beta mean
- **Derivative**: $\frac{d\mu}{d\eta} = \mu(1-\mu)$
- **Numerical safeguards**:
  - Clamp $\mu \in [10^{-10}, 1-10^{-10}]$ before applying link
  - Clamp $\eta \in [-30, 30]$ to prevent overflow in inverse

1.7.4 Chain Rule for Link Functions

When modeling parameter θ with link function g, we have $\eta = g(\theta)$. The score and Fisher information transform as:

$$\frac{\partial \ell}{\partial \eta} = \frac{d\theta}{d\eta} \cdot \frac{\partial \ell}{\partial \theta}$$

$$I_\eta = \left(\frac{d\theta}{d\eta}\right)^2 I_\theta$$

For log link ($\eta = \log(\theta)$):

$$\frac{d\theta}{d\eta} = \theta, \quad \frac{\partial \ell}{\partial \eta} = \theta \cdot \frac{\partial \ell}{\partial \theta}, \quad I_\eta = \theta^2 I_\theta$$

For logit link ($\eta = \text{logit}(\theta)$):

$$\frac{d\theta}{d\eta} = \theta(1-\theta), \quad \frac{\partial \ell}{\partial \eta} = \theta(1-\theta) \cdot \frac{\partial \ell}{\partial \theta}, \quad I_\eta = [\theta(1-\theta)]^2 I_\theta$$

For identity link ($\eta = \theta$):

$$\frac{d\theta}{d\eta} = 1, \quad \frac{\partial \ell}{\partial \eta} = \frac{\partial \ell}{\partial \theta}, \quad I_\eta = I_\theta$$

1.7.5 Working Response and IRLS

The working response in the IRLS algorithm is:

$$z = \eta + \frac{u}{w}$$

where $u = \frac{\partial \ell}{\partial \eta}$ and $w = I_\eta$.

For a Poisson model with log link: - Current: $\eta = \log(\hat{\mu})$ - Score: $u = y - \hat{\mu}$ - Weight: $w = \hat{\mu}$ - Working response: $z = \log(\hat{\mu}) + \frac{y-\hat{\mu}}{\hat{\mu}}$

For a Binomial model with logit link: - Current: $\eta = \text{logit}(\hat{\mu})$ - Score: $u = y - n\hat{\mu}$ - Weight: $w = n\hat{\mu}(1-\hat{\mu})$ - Working response: $z = \text{logit}(\hat{\mu}) + \frac{y-n\hat{\mu}}{n\hat{\mu}(1-\hat{\mu})}$

The PWLS problem then becomes:

$$\hat{\beta}^{(t+1)} = \arg\min_\beta \sum_i w_i (z_i - x_i^T \beta)^2 + \sum_j \lambda_j \beta^T S_j \beta$$

1.8 8. GCV Gradient for Smoothing Parameter Optimization

The GCV score is minimized using L-BFGS optimization. The gradient with respect to $\log(\lambda_j)$ requires careful derivation.

1.8.1 Setup

Let $V = (X^T W X + \sum_j \lambda_j S_j)^{-1}$ and $\hat{\beta}(\lambda) = V X^T W z$.

1.8.2 Key Derivatives

Derivative of β with respect to λ_j:

$$\frac{\partial \hat{\beta}}{\partial \lambda_j} = -V S_j \hat{\beta}$$

Derivative of RSS:

$$\frac{\partial \text{RSS}}{\partial \lambda_j} = 2(X^T W r)^T V S_j \hat{\beta}$$

where $r = z - X\hat{\beta}$ are the residuals.

Derivative of EDF:

$$\frac{\partial \text{EDF}}{\partial \lambda_j} = -\text{tr}(V S_j V X^T W X)$$

1.8.3 Full GCV Gradient

Using the quotient rule on $\text{GCV} = \frac{n \cdot \text{RSS}}{(n - \text{EDF})^2}$:

$$\frac{\partial \text{GCV}}{\partial \lambda_j} = \frac{n}{(n - \text{EDF})^3} \left[\frac{\partial \text{RSS}}{\partial \lambda_j}(n - \text{EDF}) + 2 \cdot \text{RSS} \cdot \frac{\partial \text{EDF}}{\partial \lambda_j} \right]$$

1.8.4 Chain Rule for Log-Space

Since we optimize in log-space:

$$\frac{\partial \text{GCV}}{\partial \log(\lambda_j)} = \lambda_j \cdot \frac{\partial \text{GCV}}{\partial \lambda_j}$$

This ensures $\lambda_j > 0$ without explicit constraints.

1.8.5 8.1 Efficient PWLS Solution via Cholesky Decomposition

The penalized weighted least squares problem:

$$\hat{\beta} = \arg\min_{\beta} \|W^{1/2}(z - X\beta)\|^2 + \sum_j \lambda_j \beta^T S_j \beta$$

has the normal equations:

$$(X^T W X + S_\lambda)\hat{\beta} = X^T W z$$

where $S_\lambda = \sum_j \lambda_j S_j$.

Key observation: $X^T W X + S_\lambda$ is symmetric positive definite (SPD) when S_j are positive semi-definite and W has positive diagonal entries.

1.8.5.1 Cholesky Approach

1. Form the weighted design matrix: $X_w = W^{1/2}X$
2. Form the coefficient matrix: $A = X_w^T X_w + S_\lambda$
3. Compute Cholesky decomposition: $A = LL^T$ where L is lower triangular
4. Solve $Ly = X^T W z$ for y (forward substitution)
5. Solve $L^T \hat{\beta} = y$ for $\hat{\beta}$ (backward substitution)

Advantages over LU decomposition: - Exploits SPD structure: $O(p^3/3)$ instead of $O(2p^3/3)$ - Numerically stable for well-conditioned systems - Automatic failure detection: Cholesky fails if matrix is not positive definite

Fallback: If Cholesky fails (due to numerical issues or near-singularity), fall back to LU decomposition with partial pivoting.

1.8.5.2 Covariance Matrix

The covariance matrix for $\hat{\beta}$ is:

$$\mathrm{Cov}(\hat{\beta}) = (X^T W X + S_\lambda)^{-1} = A^{-1}$$

With Cholesky $A = LL^T$:

$$A^{-1} = (L^T)^{-1} L^{-1}$$

We compute L^{-1} via triangular inversion, then form $(L^{-1})^T L^{-1}$.

1.8.5.3 Gradient Computation for GCV

For GCV optimization, we need $\frac{\partial \mathrm{EDF}}{\partial \lambda_j}$:

$$\frac{\partial \mathrm{EDF}}{\partial \lambda_j} = -\mathrm{tr}(V S_j V X^T W X)$$

where $V = A^{-1}$.

Exploiting block structure: When S_j is block-sparse, only compute:

$$\mathrm{tr}(V[i:i+k,:] S_{\mathrm{block}} V[:,i:i+k] X^T W X)$$

This avoids forming the full $p \times p$ products when $S_{\mathrm{block}} \ll p$.

1.8.6 8.2 REML / Laplace-Approximate Marginal Likelihood

GCV is one option for picking the smoothing parameters; the default `criterion` in `FitConfig` is `Reml` (see `src/fitting/mod.rs:43-54`). REML in this setting is the Laplace-approximate marginal likelihood (LAML) of Wood (2011), evaluated at the working PWLS step. The selector is exposed as a three-variant enum:

```
pub enum SmoothingCriterion { Gcv, Reml /* default */, FellnerSchall }
```

`Gcv` minimizes the score of §8 via L-BFGS on $\log \lambda$. `Reml` minimizes $-V_r$ (below) via L-BFGS on $\log \lambda$. `FellnerSchall` targets the same $-V_r$ via a deterministic multiplicative fixed-point (§8.3).

1.8.6.1 The LAML objective

Let $S_\lambda = \sum_j \lambda_j S_j$. The working-model LAML negative log-marginal likelihood is

$$-V_r(\lambda) = \tfrac{1}{2}\,\log\left|X^T W X + S_\lambda\right| \;-\; \tfrac{1}{2}\,\log\left|S_\lambda\right|_+ \;+\; \text{Gaussian PWLS terms (data, } \sigma)$$

where $|\cdot|_+$ denotes the **pseudo-determinant** — the product of strictly positive eigenvalues — required because S_λ is rank-deficient (its null space contains the unpenalized polynomial directions: constants for order-1 penalties, lines for order-2, etc., plus the unpenalized intercept and linear columns).

Source: `src/fitting/solver.rs:170-250` (`RemlCost`).

1.8.6.2 Pseudo-determinant via eigendecomposition

S_λ is real symmetric PSD. Decompose

$$S_\lambda = U\,\mathrm{diag}(d_1, \dots, d_p)\,U^T, \qquad d_1 \geq \cdots \geq d_p \geq 0.$$

Declare eigenvalue d_i numerically zero when

$$d_i \;<\; \tau \cdot d_1, \qquad \tau = \texttt{REML\_RANK\_TOL\_EPS} = 10^{-8}$$

(`src/fitting/solver.rs:23`). Then

$$\log|S_\lambda|_+ = \sum_{d_i > \tau d_1} \log d_i, \qquad S_\lambda^+ = U\,\mathrm{diag}(d_i^{-1}\mathbf{1}\{d_i > \tau d_1\})\,U^T.$$

Source: `src/fitting/solver.rs:547-595` (`penalty_eigen`).

1.8.6.3 REML gradient

From Wood (2011, §3) the gradient with respect to $\log\lambda_j$ is

$$\frac{\partial(-V_r)}{\partial \log\lambda_j} = \tfrac{1}{2}\,\lambda_j\,\mathrm{tr}\!\big(V\,S_j\big) \;-\; \tfrac{1}{2}\,\lambda_j\,\mathrm{tr}\!\big(S_\lambda^+\,S_j\big) \;-\; \tfrac{1}{2}\,\lambda_j\,\hat\beta^T S_j\,\hat\beta/\hat\sigma^2$$

with $V = (X^T W X + S_\lambda)^{-1}$. The implementation evaluates this analytically and feeds it to L-BFGS; final $\log\lambda$ values are clamped to $[-\texttt{LOG\_LAMBDA\_CLAMP}, \texttt{LOG\_LAMBDA\_CLAMP}] = [-30, 30]$ to keep $\lambda \in [e^{-30}, e^{30}]$.

Source: `src/fitting/solver.rs:214-303`.

1.8.7 8.3 Fellner–Schall Multiplicative Fixed Point

Wood & Fasiolo (2017) show that for the LAML target, the update

$$\lambda_j^{(t+1)} \;=\; \lambda_j^{(t)} \cdot \frac{\mathrm{tr}\!\big(S_\lambda^+\,S_j\big) - \mathrm{tr}\!\big(V\,S_j\big)}{\hat\beta^T S_j\,\hat\beta}$$

is **monotone non-increasing** in $-V_r$ under mild conditions, has no line search, and converges geometrically near a minimum. The implementation lives at `src/fitting/solver.rs:377-470` and applies three safeguards:

Safeguard	Constant	Value	Role
Relative floor on the numerator $\mathrm{tr}(S_\lambda^+ S_j) - \mathrm{tr}(V S_j)$	`FS_NUMERATOR_FLOOR_REL`	10^{-12}	Prevents tiny / occasionally negative numerators from collapsing the update.
Floor on the denominator $\hat\beta^T S_j \hat\beta$	`FS_DENOMINATOR_FLOOR`	10^{-12}	Avoids division by zero when $\hat\beta$ lies in the null space of S_j.
Per-iteration clamp on $\log\lambda_j$	`LOG_LAMBDA_CLAMP`	± 30	Keeps λ in $[e^{-30}, e^{30}]$, matching the REML L-BFGS clamp.

Loop control: at most `FS_MAX_ITERS = 50` iterations; converged when $\max_j |\log\lambda_j^{(t+1)} - \log\lambda_j^{(t)}| < \texttt{FS\_TOL} = 10^{-3}$. Source: `src/fitting/solver.rs:30-43`.

1.8.7.1 When to pick which criterion

- **GCV**: classical choice, computationally cheap; tends to undersmooth at moderate n and is more prone to multiple local minima on the GCV surface (Reiss & Ogden 2009).
- **REML** (default): preferred for stability; the Laplace-approximate marginal likelihood is asymptotically equivalent to true REML for the working PWLS model, and tends to undersmooth less than GCV at moderate sample sizes.
- **FellnerSchall**: same target as REML but with a deterministic multiplicative update — no L-BFGS, no line search. Fast and well-behaved for well-conditioned problems; can stall if the numerator drifts to its floor.

1.9 9. Mean-Variance Relationships

A key feature distinguishing different distributions is how variance relates to the mean. This determines which distribution is appropriate for different data structures.

Distribution	Mean	Variance	Relationship
Gaussian	μ	σ^2	Constant (homoskedastic)
Poisson	μ	μ	Equidispersion: Var = Mean
Gamma	μ	$\mu^2\sigma^2$	Proportional to μ^2 (constant CV)
Negative Binomial	μ	$\mu + \sigma\mu^2$	Overdispersion relative to Poisson
Binomial	$n\mu$	$n\mu(1-\mu)$	Quadratic in mean, max at $\mu = 0.5$
Beta	μ	$\frac{\mu(1-\mu)}{1+\phi}$	Max at $\mu = 0.5$, decreases with ϕ
Student-t	μ	$\frac{\nu\sigma^2}{\nu-2}$	Heavier tails than Gaussian

1.9.1 Choosing a Distribution

Count data: - Equidispersed: Poisson - Overdispersed (variance > mean): Negative Binomial - Known number of trials: Binomial

Continuous positive: - Constant coefficient of variation: Gamma - Heavy tails: Student-t with ν small - Heteroskedastic with mean: Gamma or lognormal (not implemented)

Proportions/rates in $(0, 1)$: - Beta (continuous) - Binomial (discrete counts / n trials)

Continuous unbounded: - Light tails, constant variance: Gaussian - Heavy tails: Student-t

1.9.2 Overdispersion

Overdispersion occurs when the observed variance exceeds what the distribution predicts from the mean alone.

For count data: - If $\text{Var}(Y) > \mu$: Use Negative Binomial instead of Poisson - The parameter σ in NB quantifies the overdispersion: as $\sigma \to 0$, NB $\to$ Poisson

For continuous data: - Gamma allows variance to scale with μ^2 - Student-t allows heavier tails (more extreme values) than Gaussian

---

1.10 10. Convergence and Initialization

1.10.1 Initialization Strategy

The RS algorithm requires starting values for all parameters. Default initialization:

1. μ:
   - Gaussian/Student-t: $\bar{y}$ (sample mean)
   - Poisson/Gamma/NB: $\bar{y}$ (then apply log link)
   - Binomial: $\bar{y}/n$ (sample proportion)
   - Beta: $\bar{y}$ (sample mean, clamped to $(0.1, 0.9)$)
2. σ:
   - Gaussian/Student-t: s_y (sample std dev)
   - Gamma/NB/Beta: 1.0 (or estimated from sample CV)
3. ν (Student-t): 5.0
4. ϕ (Beta): 1.0

1.10.2 Convergence Criterion

The algorithm stops when:

$$\max_k \|\beta_k^{(t+1)} - \beta_k^{(t)}\|_1 < \epsilon$$

where k indexes parameters (μ, σ, etc.) and $\epsilon = 10^{-3}$ by default.

Maximum iterations: 200 (default)

1.10.3 Convergence Issues

Non-convergence can occur due to: 1. **Poor initialization**: Try better starting values, especially for ν in Student-t 2. **Model misspecification**: Wrong distribution family for the data 3. **Multicollinearity**: Highly correlated predictors (check condition number of design matrix) 4. **Insufficient smoothing**: Extremely small λ can cause instability 5. **Numerical issues**: Extreme data values causing overflow/underflow

Diagnostics: - Monitor deviance: should decrease monotonically - Check for NaN/Inf in coefficients or fitted values - Inspect condition number of $(X^T W X + \lambda S)$

1.11 11. Model Selection and Diagnostics

1.11.1 11.1 Information Criteria

Akaike Information Criterion (AIC):

$$\text{AIC} = -2\ell(\hat{\theta}) + 2k$$

where $\ell(\hat{\theta})$ is the maximized log-likelihood and k is the number of parameters.

For GAMLSS with smoothing:

$$k = \sum_{j=1}^{P} \text{EDF}_j$$

where P is the number of distribution parameters and EDF_j accounts for the effective degrees of freedom of smooth terms.

Bayesian Information Criterion (BIC):

$$\text{BIC} = -2\ell(\hat{\theta}) + k\log(n)$$

BIC penalizes complexity more heavily than AIC for $n > 7$.

1.11.2 11.2 Deviance

The **deviance** measures lack of fit:

$$D = 2[\ell_{\text{saturated}} - \ell_{\text{fitted}}]$$

where $\ell_{\text{saturated}}$ is the log-likelihood of a saturated model (perfect fit).

For nested models, the deviance difference follows approximately χ^2 distribution:

$$D_1 - D_2 \sim \chi^2_{\text{EDF}_2 - \text{EDF}_1}$$

1.11.3 11.3 Residuals

Quantile (randomized quantile) residuals:

$$r_i = \Phi^{-1}(F(y_i|\hat{\theta}_i))$$

where F is the fitted CDF and Φ^{-1} is the inverse standard normal CDF.

If the model is correct, $r_i \sim N(0,1)$.

Deviance residuals:

$$r_i^{(D)} = \text{sign}(y_i - \hat{\mu}_i)\sqrt{d_i}$$

where d_i is the contribution of observation i to the total deviance.

Pearson residuals:

$$r_i^{(P)} = \frac{y_i - \hat{\mu}_i}{\sqrt{\widehat{\text{Var}}(Y_i)}}$$

1.11.4 11.4 Worm Plots

For each parameter θ_k, plot quantile residuals against fitted quantiles: - X-axis: Normal quantiles $\Phi^{-1}((i - 0.5)/n)$ - Y-axis: Sorted quantile residuals

If the model is correct, points should lie on a horizontal line at zero. Systematic deviations indicate: - U-shape: Underdispersion - Inverse U-shape: Overdispersion - S-shape: Skewness issues - Linear trend: Location shift

1.11.5 11.5 Q-Q Plots

Plot theoretical quantiles against sample quantiles of residuals. Should be approximately linear if residuals are normal.

1.11.6 11.6 Goodness of Link Test

To test if link g is appropriate, fit augmented model:

$$g(\theta) = X\beta + \gamma[g(\theta)]^2$$

Test $H_0 : \gamma = 0$ using likelihood ratio test.

1.12 12. Code Map

A reverse index from mathematical concept to the canonical implementation, for contributors jumping between formula and source.

Concept (this doc)	Source location
Backfitting outer loop (§4.1)	`src/fitting/mod.rs:223-263`
P-IRLS inner step (§4.2)	`src/fitting/scoring.rs:40-180`
`MIN_WEIGHT`, `MAX_STEP` (§4.4)	`src/fitting/scoring.rs:28,32`
`MAX_ETA`, `MIN_ETA`, `MIN_POSITIVE` (§4.4)	`src/distributions/links.rs:8-12`
`IdentityLink`, `LogLink`, `LogitLink` (§7)	`src/distributions/links.rs:24-59`
Gaussian derivatives (§1.1)	`src/distributions/gaussian.rs`
Student-t derivatives (§1.2)	`src/distributions/student_t.rs`
Poisson derivatives (§1.3)	`src/distributions/poisson.rs`
Gamma derivatives (§1.4)	`src/distributions/gamma.rs`
Negative Binomial derivatives (§1.5)	`src/distributions/negative_binomial.rs`
Beta derivatives (§1.6)	`src/distributions/beta.rs`
Binomial derivatives (§1.7)	`src/distributions/binomial.rs`
Digamma / trigamma batched (§2)	`src/math.rs`
Distribution trait (§1)	`src/distributions/mod.rs`
B-spline basis & knots (§5.0)	`src/splines.rs:42-160`
Sum-to-zero Householder Z (§5.1)	`src/splines.rs:1-39`
Difference penalty matrix (§5)	`src/splines.rs:162-196`
Tensor product assembly (§5)	`src/fitting/assembler.rs:44-102`
Random effect assembly (§5)	`src/fitting/assembler.rs:103-126`
Block-sparse penalty (§5.1)	`src/fitting/assembler.rs:128-187`, `src/types/newtypes.rs`
PWLS / Cholesky solve (§8.1)	`src/fitting/solver.rs:466-545`
EDF (§6)	`src/fitting/solver.rs:514-530`
GCV cost and gradient (§8)	`src/fitting/solver.rs:67-160`
GCV L-BFGS driver	`src/fitting/solver.rs:290-373`
REML / LAML cost and gradient (§8.2)	`src/fitting/solver.rs:170-303`
Penalty pseudo-determinant / pseudo-inverse (§8.2)	`src/fitting/solver.rs:547-595` (`penalty_eigen`)
Fellner–Schall iteration (§8.3)	`src/fitting/solver.rs:377-470`
`SmoothingCriterion` enum (§8.2)	`src/fitting/mod.rs:46-54`
Defaults (`max_iterations=200`, `tolerance=1e-3`)	`src/fitting/mod.rs:31-32`
Diagnostics (AIC/BIC/EDF/residuals, §11)	`src/fitting/diagnostics.rs`

1.13 13. Notation Glossary

Symbols used throughout this document.

Symbol	Meaning
y, Y	Response value / random variable
n	Number of observations
p	Total number of coefficients across all terms; also (in §5.0) B-spline degree
P	Number of distribution parameters (e.g. $P = 2$ for Gaussian)
θ_k	k-th distribution parameter (μ, σ, ν, ϕ)

Symbol	Meaning
μ, σ, ν, ϕ	Location, scale, degrees of freedom (Student-t), precision (Beta)
g, g^{-1}	Link function and its inverse
$\eta = g(\theta)$	Linear predictor for a distribution parameter
X	Design matrix ($n \times p$); collects intercept, linear, and smooth basis columns
β	Coefficient vector ($p \times 1$)
W	Diagonal weight matrix; $W_{ii} = w_i$ = Fisher info for observation i
$u_i = \partial\ell/\partial\eta_i$	Score (first derivative of log-likelihood on η-scale)
$w_i = -\mathbb{E}[\partial^2\ell/\partial\eta_i^2]$	Fisher information on η-scale
$z = \eta + u/w$	Working response
S_j	j-th penalty matrix (one per smooth term direction)
λ_j	Smoothing parameter for S_j
$S_\lambda = \sum_j \lambda_j S_j$	Aggregate penalty matrix
$V = (X^T W X + S_\lambda)^{-1}$	PWLS coefficient covariance
$H = XVX^T W$	Hat matrix (projection onto fitted values)
$\mathrm{EDF} = \mathrm{tr}(H) = \mathrm{tr}(VX^T WX)$	Effective degrees of freedom
$\mathrm{RSS} = (z - X\hat{\beta})^T W (z - X\hat{\beta})$	Weighted residual sum of squares
$\psi(x), \psi'(x)$	Digamma, trigamma
$\|A\|_+$	Pseudo-determinant of PSD matrix A (product of strictly positive eigenvalues)
A^+	Moore–Penrose pseudo-inverse of A
$\otimes$	Kronecker product
$\odot$ (in §5)	Row-wise Kronecker product (one row of $B_1 \otimes B_2$ per row index)
$\mathbf{1}_k$	Column vector of k ones
I_k	$k \times k$ identity matrix

1.14 References

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